

Challenge 3: Caccetta–Häggkvist Conjecture

Definition 1 (Out-degree). Let D be a simple digraph with vertex set $V(D)$ and edge set $E(D)$. The *out-degree* $d^+(v)$ of a vertex v is the number of edges $(v, w) \in E(D)$, and the *minimum out-degree* of D is the smallest out-degree of a vertex of D .

Definition 2 (Directed cycle). A *directed cycle of length $\ell \geq 2$* in a simple digraph D is a sequence $v_0, \dots, v_{\ell-1}$ of distinct vertices such that $(v_i, v_{i+1}) \in E(D)$ for every i , where the indices are taken modulo ℓ .

Conjecture 1.1 (Caccetta–Häggkvist Conjecture [1]). Let $r \in \mathbb{N}$ with $r \geq 1$. Every simple directed graph on n vertices with minimum out-degree at least r has a cycle of length at most $\lceil n/r \rceil$.

For $r = 1$ the statement is elementary. The conjecture was proved for $r = 2$ by Caccetta and Häggkvist [1], for $r = 3$ by Hamidoune [2], and for $r = 4$ and $r = 5$ by Hoàng and Reed [3]. Shen proved it for every r under the additional assumption $n \geq 2r^2 - 3r + 1$ [4].

References

- [1] L. Caccetta and R. Häggkvist. “On minimal digraphs with given girth”. In: *Proceedings of the Ninth Southeastern Conference on Combinatorics, Graph Theory, and Computing (Boca Raton, 1978)*. Vol. 21. Congressus Numerantium. Utilitas Mathematica, 1978, pp. 181–187.
- [2] Y. O. Hamidoune. “A note on minimal directed graphs with given girth”. In: *Journal of Combinatorial Theory, Series B* 43.3 (1987), pp. 343–348.
- [3] C. T. Hoàng and B. Reed. “A note on short cycles in digraphs”. In: *Discrete Mathematics* 66.1–2 (1987), pp. 103–107.
- [4] J. Shen. “On the girth of digraphs”. In: *Discrete Mathematics* 211.1–3 (2000), pp. 167–181.